MCO — METACHRONAL CASCADE ORNITHOPTER

Wing Manufacturing Document

MCO-1D Drone Prototype

S1223 Airfoil Specification · Universal LW-PLA Rib Design · SolidWorks Parameters · Bill of Materials · Assembly Instructions

Design philosophy	One universal rib design at 120mm chord. Membrane trimmed to each wing's planform. Ribs left full-length as internal formers. Aerodynamic shape defined by the membrane, not the rib edge.
Performance	Approximately 1-2% efficiency reduction versus the fully optimised 12-configuration design. Acceptable for prototype. Recoverable on second build.
Build count	110 identical ribs across all 8 wings. One slicer setup. One print run of approximately 15-22 hours. No trimming of ribs required.
Document structure	Chapter 1: Airfoil specification and properties. Chapter 2: SolidWorks rib parameters and feature tree. Chapter 3: Coordinate file for profile import. Chapter 4: Bill of materials. Chapter 5: Assembly instructions step by step. Blueprint: Annotated cross-section schematic.

CHAPTER 1 — THE S1223 AIRFOIL — SPECIFICATION AND PROPERTIES

1.1 Why S1223

The Selig S1223 was designed specifically for high-lift low-Reynolds-number applications. It is the correct choice for the MCO-1D drone operating at chord Reynolds numbers of 3,500 to 26,000 across the cascade. It provides the highest maximum lift coefficient of any section in its class while retaining a soft stall characteristic — stall progresses gradually from root to tip rather than snapping off abruptly. This soft stall behaviour is essential for a flapping wing that operates at high angles of attack on every power stroke.

1.2 Key Geometric Properties

Designation	Selig S1223
Source	UIUC Airfoil Database — s1223.dat
Maximum thickness	12.14% chord at 30.08% chord position
Maximum camber	8.68% chord at 47.32% chord position
Leading edge radius	1.56% chord
Trailing edge	Blunted at 97% chord — 0.80mm flat close (printability)
At 120mm chord:	
Max thickness	14.57mm at 36.1mm from leading edge
Max camber	10.42mm at 56.8mm from leading edge
LE radius	1.87mm radius
TE close point	116.4mm from LE (97% chord)
TE close width	0.80mm flat vertical close
Design Reynolds number	100,000 to 300,000 (original design range)

MCO operating Re	3,500 to 26,600 — below design range; S1223 still outperforms alternatives at these Re due to high camber maintaining attached flow with LEV assistance
CL max (experimental)	2.4 to 2.6 at Re 100,000 (with VGs and covert flaps)
Best L/D AoA	8 to 10 degrees — used for glide and WING-RANGE modes
Power stroke AoA	35 to 42 degrees — deep post-stall, LEV-dominated
Stall character	Soft stall — root first, gradual progression outboard

1.3 The Trailing Edge Modification

The original S1223 has a trailing edge of approximately 0.25% chord thickness at the tip — 0.30mm at 120mm chord. This cannot be printed on a 0.4mm nozzle. The modification truncates the profile at 97% chord (116.4mm from the LE) and closes with a vertical 0.80mm flat face. This is standard practice in model aircraft construction and has been shown to have negligible aerodynamic effect below Reynolds number 200,000. The blunt TE also provides a clean, consistent attachment point for the trailing edge carbon strip and membrane.

The 97% chord truncation is already incorporated into all coordinate tables in this document. No further modification is required in SolidWorks. The coordinate file terminates at x = 116.4mm, not 120mm.

CHAPTER 2 — SOLIDWORKS RIB PARAMETERS

2.1 Global Variables

Enter these in SolidWorks via Tools → Equations → Global Variables. Every feature dimension references these variables. The rib is fully defined by these values with no additional inputs.

Variable	Value	Unit	Purpose
Chord_mm	120.00	mm	Master chord length. Do not change for the universal rib.
Rib_thickness	2.00	mm	Extrusion depth. Constant for all ribs.
LE_bore_dia	2.10	mm	LE spar clearance bore. Fits 2.0mm OD spar.
Sec_spar_dia	3.10	mm	Secondary spar notch diameter. Fits 2.0mm OD rod.
Sec_spar_x	72.00	mm	Secondary spar X position (60% of 120mm chord).
TE_x	116.40	mm	Trailing edge truncation point (97% chord).
TE_close	0.80	mm	Blunt TE close width.
TE_chamfer_run	0.40	mm	TE chamfer distance.
TE_chamfer_angle	30.00	deg	TE chamfer angle.
LE_solid_x	18.00	mm	LE solid zone boundary (15% of 120mm).
Mid_zone_end_x	96.00	mm	Mid zone end (80% of 120mm).
Flap_row1_x	45.00	mm	Film flap row 1 forward edge (37.5% chord).
Flap_row2_x	63.00	mm	Film flap row 2 forward edge (52.5% chord).
Flap_row3_x	81.00	mm	Film flap row 3 forward edge (67.5% chord).
Flap_width	9.60	mm	Film flap chord width (8% of 120mm).
Flap_recess_depth	0.30	mm	Film flap bond ledge recess depth.
VG_x	20.40	mm	VG recess centre X (17% of 120mm chord).
VG_recess_depth	0.20	mm	VG recess depth.
VG_recess_width	2.00	mm	VG recess chord width.

Variable	Value	Unit	Purpose
VG_dimple_dia	0.30	mm	VG alignment dimple diameter.
VG_dimple_depth	0.15	mm	VG alignment dimple depth.
Bore_fillet_r	0.30	mm	Fillet radius on LE bore edges.

2.2 Feature Tree — Step by Step

Build features in this exact order. Each step references the global variables defined above.

STEP 1 — Create New Part

File → New → Part. Set document units to MILLIMETRES (Tools → Options → Document Properties → Units). Enter all Global Variables from Section 2.1.

STEP 2 — Import Profile (Base Sketch)

Insert → Curve → Curve Through XYZ Points. Browse to the coordinate file from Chapter 3. SolidWorks imports 61 points and fits a spline. The result is a closed profile on the Front Plane with LE at the origin.

STEP 3 — Base Extrusion

Select the closed profile. Features → Extruded Boss/Base. Depth = 'Rib_thickness' = 2.00mm. Direction: along the Z axis. Result: 2.00mm thick S1223 solid.

STEP 4 — LE Spar Bore

Sketch2 on front face of rib. Draw circle: diameter = 'LE_bore_dia' = 2.10mm. Centre: X = 1.80mm, Y = 1.85mm (above chord line). Features → Extruded Cut → Through All Both Directions. Then add Fillet: radius = 'Bore_fillet_r' = 0.30mm on both bore entry edges.

STEP 5 — Secondary Spar Notch

Sketch3 on TOP face (upper surface). Draw semicircle: diameter = 'Sec_spar_dia' = 3.10mm, open end facing upward. Centre X = 'Sec_spar_x' = 72.00mm. Centre Y: use Pierce constraint to the upper surface spline (approximately 11.60mm above chord line at 60% chord). Cut: 2.00mm deep downward through rib.

STEP 6 — Trailing Edge Chamfer

Select both TE edges (upper and lower faces at x = 116.4mm). Insert → Features → Chamfer. Distance = 'TE_chamfer_run' = 0.40mm. Angle = 'TE_chamfer_angle' = 30°. Apply to both edges.

STEP 7 — Film Flap Ledge Row 1

Sketch4 on upper face. Rectangle: chord-width = 'Flap_width' = 9.60mm, span-width = 'Rib_thickness' = 2.00mm. Forward edge at X = 'Flap_row1_x' = 45.00mm. Use Pierce constraint to position Y on upper surface spline. Cut: 'Flap_recess_depth' = 0.30mm deep perpendicular to upper surface.

STEP 8 — Film Flap Ledge Row 2

Identical to Step 7. Forward edge at X = 'Flap_row2_x' = 63.00mm.

STEP 9 — Film Flap Ledge Row 3

Identical to Steps 7-8. Forward edge at X = 'Flap_row3_x' = 81.00mm.

STEP 10 — VG Recess

Sketch7 on upper face. Rectangle: chord-width = 'VG_recess_width' = 2.00mm, span-width = 2.00mm. Centre at X = 'VG_x' = 20.40mm. Pierce to upper surface spline for Y. Cut: 'VG_recess_depth' = 0.20mm deep.

STEP 11 — VG Alignment Dimple

Sketch8 on the floor of the VG recess. Draw point at centre. Insert → Features → Revolved Cut. Profile: right triangle with base 0.15mm, height 0.15mm, 60° included angle. Revolve 360°. Result: conical dimple 0.30mm dia × 0.15mm deep at VG recess centre.

STEP 12 — Split Bodies for Infill Zones

Insert → Features → Split. Trim tool 1: Reference Plane offset from Front Plane at X = 'LE_solid_x' = 18.00mm (15% chord). Trim tool 2: Reference Plane at X = 'Mid_zone_end_x' = 96.00mm (80% chord). Result: three separate bodies. Name them: Body1-LE-Solid, Body2-Mid-Gyroid, Body3-TE-Gyroid.

STEP 13 — Save and Export

File → Save As → MCO_UniversalRib_120mm.SLDPRT. Then export as STL: File → Save As → STL. Set resolution to Fine. Export all three bodies as separate STL files: MCO_Rib_Body1_LESolid.stl, MCO_Rib_Body2_MidGyroid.stl, MCO_Rib_Body3_TEGyroid.stl.

2.3 Critical Feature Positions Summary

Feature	Position	Depth / Height	Dimensions
LE spar bore centre	X = 1.80mm, Y = 1.85mm	Through all	2.10mm dia + 0.30mm edge fillet
Secondary spar notch	X = 72.00mm, Y = upper surface	2.00mm deep	3.10mm dia semicircle
LE solid zone boundary	X = 18.00mm (15%c)	Split plane	Infill zone separator
Mid/TE zone boundary	X = 96.00mm (80%c)	Split plane	Infill zone separator
Film flap ledge R1	X = 45.00mm (37.5%c)	0.30mm deep	9.60mm wide × 2.00mm span
Film flap ledge R2	X = 63.00mm (52.5%c)	0.30mm deep	9.60mm wide × 2.00mm span
Film flap ledge R3	X = 81.00mm (67.5%c)	0.30mm deep	9.60mm wide × 2.00mm span
VG recess centre	X = 20.40mm (17%c)	0.20mm deep	2.00mm × 2.00mm + dimple
TE truncation	X = 116.40mm (97%c)	0.80mm close	Blunt TE close
TE chamfer both edges	X = 116.40mm	0.40mm run, 30°	Upper and lower faces

CHAPTER 3 — COORDINATE FILE FOR SOLIDWORKS IMPORT

Save the following as a plain text file named MCO_S1223_120mm.sldcrv (or .txt). Three columns: X, Y, Z. Space-separated. No headers. Import via Insert → Curve → Curve Through XYZ Points. All values in millimetres. Z = 0.000 throughout. 61 points form the complete closed profile including the blunt TE close.

The upper surface runs from the LE (0,0) to the TE (116.4, 0). The TE close line adds one intermediate point at (116.4, -0.8). The lower surface returns from the TE to the LE. SolidWorks fits a single closed spline through all 61 points in order.

UPPER SURFACE — 30 points (LE to TE):

0.000	0.000	0.000
0.151	1.001	0.000
0.542	1.958	0.000
1.522	3.416	0.000
3.048	4.914	0.000
5.729	6.872	0.000
9.253	8.771	0.000
13.516	10.522	0.000
18.388	12.047	0.000
23.738	13.297	0.000
29.436	14.244	0.000
36.124	14.861	0.000
41.326	15.108	0.000
47.244	14.996	0.000
52.963	14.544	0.000
58.354	13.791	0.000
63.293	12.791	0.000
67.629	11.597	0.000
71.337	10.273	0.000

74.141	8.896	0.000
78.918	7.291	0.000
84.101	5.616	0.000
89.597	4.049	0.000
95.174	2.686	0.000
100.533	1.622	0.000
105.640	0.878	0.000
110.054	0.415	0.000
113.975	0.137	0.000
116.622	0.002	0.000
116.400	0.000	0.000

TRAILING EDGE CLOSE — 2 points:

116.400	0.000	0.000
116.400	-0.800	0.000

LOWER SURFACE — 29 points (TE back to LE):

116.400	-0.800	0.000
116.622	-0.036	0.000
113.975	-0.143	0.000
110.054	-0.393	0.000
105.640	-0.623	0.000
100.533	-0.798	0.000
95.174	-0.917	0.000
89.597	-0.979	0.000
84.101	-0.972	0.000
78.918	-0.878	0.000
74.141	-0.769	0.000
71.337	-0.630	0.000

67.629	-0.498	0.000
63.293	-0.357	0.000
58.354	-0.214	0.000
52.963	-0.099	0.000
47.244	-0.009	0.000
41.326	0.040	0.000
36.124	0.076	0.000
29.436	0.075	0.000
23.738	0.042	0.000
18.388	-0.017	0.000
13.516	-0.088	0.000
9.253	-0.194	0.000
5.729	-0.318	0.000
3.048	-0.396	0.000
1.522	-0.339	0.000
0.542	-0.206	0.000
0.151	-0.092	0.000
0.000	0.000	0.000

Once imported the curve should appear as a smooth closed teardrop shape with the LE at the origin, chord running right along the X axis, and the camber curving upward. If the curve appears distorted, check that document units are set to millimetres before importing.

CHAPTER 4 — 3D PRINT SETTINGS

Material	ColorFabb LW-PLA or Spectrum Ultra Foam PLA
Nozzle temperature	252°C (full activation, maximum foaming)
Bed temperature	60°C, smooth PEI sheet
Layer height	0.15mm
Nozzle diameter	0.40mm
Print speed	35 mm/s
Fan speed	80% from layer 3 onward
Extrusion multiplier	0.67 (tune ±0.02 on first test rib)
Print orientation	Rib FLAT on bed: chord horizontal, span (Z axis) vertical. This aligns print layers with aerodynamic load direction.
Supports	NONE required in this orientation
Top / bottom layers	2 solid layers
Body 1 infill	100% solid (LE nose zone 0-18mm). Set via modifier mesh in OrcaSlicer or PrusaSlicer.
Body 2 infill	12% Gyroid (mid zone 18-96mm)
Body 3 infill	8% Gyroid (TE zone 96-116.4mm)
Body 1 perimeters	4 perimeters = 1.60mm wall
Body 2 perimeters	3 perimeters = 1.20mm wall
Body 3 perimeters	2 perimeters = 0.80mm wall
Estimated time per rib	8 to 12 minutes
Total print time (110 ribs)	Approximately 15 to 22 hours
Effective density	Approximately 0.55 g/cm³ (44% of standard PLA)
Expected mass per rib	0.80 to 1.15g (adjust extrusion multiplier to achieve)

Print one test rib first. Verify mass is 0.80-1.15g. Verify LE bore accepts 2.0mm spar rod freely. Verify 2.2mm rod does NOT pass through bore. Do not batch-print 110 ribs until test rib passes all checks.

CHAPTER 5 — BILL OF MATERIALS — COMPLETE WING SYSTEM (ALL 8 WINGS)

The following covers all materials required to complete all eight wings including ribs, spars, membrane, covert flaps, VGs, fences, end plates, LE fairing strips, TE strips, and root attachment plates. Quantities include a 10% contingency allowance.

5.1 Printed Parts

Item	Quantity	Material	Notes
Universal wing rib — 120mm	110 ribs	LW-PLA (ColorFabb or equivalent)	~0.9g each = ~99g total

5.2 Carbon Fibre Structural

Item	Specification	Quantity	Notes
LE spar — W1,W2,W4 wings (6 wings)	2.0mm OD / 1.5mm ID pultruded CF tube	6 × 600mm	3.6m total + spare = order 4.5m
LE spar — W3 wings (2 wings)	2.5mm OD / 1.5mm ID pultruded CF tube	2 × 630mm	1.26m + spare = order 1.8m
Secondary spar — all 8 wings	2.0mm OD / 1.0mm ID pultruded CF tube	8 × 600mm	4.8m + spare = order 6.0m
LE fairing strip — all 8 wings	0.3mm × 6mm CFRP flat strip	8 × 600mm	4.8m + spare = order 6.0m
Trailing edge strip — all 8 wings	0.5mm × 4mm CFRP flat strip	8 × 600mm	4.8m + spare = order 6.0m
Wing fence material	0.2mm CFRP sheet	300mm × 200mm sheet	CNC or laser cut to shape
End plate material	0.5mm CFRP sheet	400mm × 300mm sheet	NACA 0008 profile, CNC cut

Item	Specification	Quantity	Notes
Root attachment plates	2.0mm CFRP sheet	8 plates 40mm × 30mm	CNC cut, M3 bolt holes

5.3 Membrane and Film Materials

Item	Specification	Quantity	Notes
Wing membrane — upper and lower	Ripstop nylon, 20 micron	2.2 m²	Grain chordwise. Buy 2.5m² to allow for cutting waste.
Film covert flaps — all rows	Ripstop nylon 20 micron OR Mylar/PET 25 micron	0.4 m²	Same material as membrane preferred. Cut into strips on assembly.

5.4 Fasteners and Adhesives

Item	Specification	Quantity	Notes
Thin CA adhesive (rib bonding)	Loctite 401 or equivalent	2 × 20ml bottles	Rib to spar bonding
Contact adhesive (membrane)	UHU Por or equivalent	1 × 100ml tin	Membrane to rib face bonding
Structural epoxy (root plates)	Araldite 2011 or Hysol EA9394	1 × 50ml syringe	Root plate to spar bonding
VG bonding film	3M 9460PC structural film adhesive	0.15 m²	Pre-cut to VG dimensions
M3 socket head cap screws	A4 stainless, 8mm length	32 screws	Root plate attachment, 4 per wing
M3 hex nuts	A4 stainless	32 nuts	Root plate attachment
M3 washers	A4 stainless	64 washers	Root plate attachment
Thread-locker	Loctite 243 medium strength	1 × 5ml bottle	All rotating fasteners

5.5 VG Array Components

Item	Specification	Quantity	Notes
VG vanes	0.3mm CFRP sheet, CNC laser cut	Per wing: approx 24 pairs	Counter-rotating delta pairs. Height 0.8% local chord (~1mm).
VG total all 8 wings	0.3mm CFRP sheet, 150mm × 150mm	1 sheet	Yields all VGs for all 8 wings

5.6 Mass Summary

Item	Mass	Notes
LW-PLA ribs (all 8 wings)	~99g	110 ribs × 0.9g average
Carbon spars (all 8 wings)	~22g	LE + secondary per wing
Membrane ripstop nylon	~24g	Upper + lower all 8 wings, 4g/m²
Film covert flaps	~4g	Three rows per wing, 20 micron film
LE fairing strips	~6g	0.3mm CFRP strip all 8 wings
TE strips	~8g	0.5mm CFRP strip all 8 wings
VG arrays	~5g	0.3mm CFRP all wings
Wing fences	~2g	0.2mm CFRP all wings
End plates	~10g	0.5mm CFRP NACA 0008 all wings
Root attachment plates	~8g	2mm CFRP 8 plates
Adhesives and fasteners	~8g	Estimated in-place mass
TOTAL 8-WING SYSTEM	~196g	Target 185-210g

CHAPTER 6 — ASSEMBLY INSTRUCTIONS — ONE COMPLETE WING

These instructions cover the assembly of one wing. Repeat for all eight. Left and right wings are mirror images — assemble the right side first, then mirror all jig settings for the left. Allow all adhesive bonds to cure fully before proceeding to the next step. Work on a flat, clean surface.

6.1 Tools Required

- Digital caliper (0.01mm resolution)
- 0.01g resolution jeweller's scale
- Flat assembly jig or flat board with dihedral angle blocks
- Impulse hammer or light pencil for natural frequency test
- Smartphone or oscilloscope for frequency measurement (apps: PhyPhox, Spectroid)
- Sharp scalpel or craft knife
- Scissors (for membrane cutting)
- Planform template cut from cardboard (trace from drawing for each wing)
- Steel rule and pencil
- Syringe applicators for CA and epoxy
- Activator spray for CA (optional but speeds cure)
- Masking tape (alignment aid)
- 220-grit sandpaper (edge finishing)

6.2 Preparation

STEP PREP-1 — Sort Ribs

Count out the required number of ribs for this wing (14 for W1, 13 for W2 and W4, 15 for W3).

Weigh each rib. Discard any outside the 0.80-1.15g range. Check LE bore with a 2.0mm rod — it must slide freely through. Check that a 2.2mm rod does NOT pass through.

Arrange in order from root to tip. Mark the root rib with a pencil dot.

STEP PREP-2 — Cut Spar to Length

Cut the LE spar tube to the wing span length: W1=550mm, W2=495mm, W3=578mm, W4=523mm.

Cut the secondary spar rod to span length minus 20mm (leaves 10mm clear each end).

Deburr both ends of all cut spars with 220-grit sandpaper wrapped around a finger.

Check both spars fit through all rib bores and notches smoothly.

STEP PREP-3 — Cut Planform Template

Cut a cardboard template of the wing planform from the wing drawing.

The template shows the exact shape the membrane will be trimmed to after bonding.

Label it with the wing number and LEFT or RIGHT.

Keep this template for the membrane trimming step.

6.3 Rib and Spar Assembly

STEP 1 — Thread Ribs onto LE Spar

Slide the root rib onto the LE spar first. The spar passes through the 2.10mm bore in the rib nose section. The rib should slide freely with no force required.

Slide all remaining ribs onto the spar in order, root to tip.

Space ribs at approximately 40mm centres in the root half of the span, 55mm centres in the tip half. Exact spacing is not critical — uniform distribution matters more than precise dimension.

Ensure the tip rib sits within 20mm of where the membrane tip edge will be trimmed. Check against the planform template.

Do not bond yet — check all ribs are oriented the same way (upper surface all same side).

STEP 2 — Set Dihedral and Bond Ribs

Place the assembly in the dihedral jig. Dihedral angles: W1=3°, W2=5°, W3=7°, W4=4°. A simple jig is a flat board with a wedge block under the tip end of the spar.

Apply thin CA to each rib-spar interface from both faces. Use a syringe applicator — capillary action draws the CA into the bore gap.

Hold each rib perpendicular to the spar for 10-15 seconds until CA grips.

Work root to tip. Allow 2 minutes full cure before moving to the next step.

Do not glue the root rib yet — it receives the root attachment plate in a later step.

STEP 3 — Install Secondary Spar

Drop the secondary spar rod into the semi-circular notches on the upper surface of all ribs. The rod should sit in the notches under gravity with zero force.

Apply thin CA to each notch contact point from above. Allow CA to wick into the notch around the rod.

The secondary spar should not be forced into any notch. If it does not drop in freely, lightly enlarge the notch with a 3mm drill or round file.

STEP 4 — Natural Frequency Test

THIS STEP IS MANDATORY BEFORE PROCEEDING.

Clamp the root end of the wing assembly in a vice or clamp (through the root attachment plate area). Wing should be horizontal, tip unsupported.

Open a frequency measurement app on a smartphone (PhyPhox Acoustic or Spectroid). Alternatively connect an accelerometer to the tip.

Tap the wingtip sharply with a fingertip. Observe the decay oscillation.

Read the frequency from the peak in the spectrum or from timing the decay cycles.

ACCEPT: 1.8 to 2.2 Hz.

If fn > 2.2 Hz (too stiff): replace LE spar with the next size down (W1/W2/W4: from 2.0mm OD try 1.5mm OD; W3: from 2.5mm OD try 2.0mm OD).

If fn < 1.8 Hz (too flexible): replace LE spar with next size up.

Record the measured fn for each wing. Wings at the same position should be within 0.1 Hz of each other.

6.4 VG Array Installation

STEP 5 — Attach VG Vanes

Install VGs before the membrane. Once the membrane is on, VG access is impossible.

VG position on the universal rib: X = 20.40mm from LE (centred in the VG recess).

Cut a strip of 3M 9460PC film adhesive to fit the VG recess footprint.

Bond VG vane pairs into the recess using the 9460PC film. Use the alignment dimple to centre the first pair.

VG height: 0.8% of local chord = approximately 1mm. VG cant angle: 18° to local flow direction. Counter-rotating pairs.

Press firmly for 30 seconds. Allow 1 hour before membrane application.

6.5 LE Fairing Strip

STEP 6 — Bond LE Fairing Strip

The 0.3mm × 6mm CFRP fairing strip bridges between rib noses along the leading edge.

Apply contact adhesive (UHU Por) to the upper surface of all rib nose sections and to one face of the fairing strip. Allow 60 seconds to become tacky.

Press the fairing strip against the leading edge of the rib array, running rib-to-rib across the full span. The strip sits on the upper surface of the rib noses and follows the spar.

Press firmly along the full length. Hold for 2 minutes.

Trim any excess at root and tip with a sharp scalpel.

This strip prevents membrane collapse between ribs at the leading edge and provides a continuous bonding surface for the upper membrane at the nose.

6.6 Membrane Application

STEP 7 — Apply Upper Membrane

Cut a membrane panel approximately 30mm larger than the wing planform in all directions. Grain direction: chordwise (parallel to the ribs).

Apply contact adhesive to the upper face of all ribs, the top of the LE fairing strip, and the top face of the TE strip (which will be bonded in the next step). Also apply adhesive to the corresponding areas of the membrane underside.

Allow 60-90 seconds. Both surfaces should be tacky but not wet.

Lay the membrane onto the wing assembly. Start at the leading edge — press the membrane firmly onto the LE fairing strip first, then work spanwise from root to tip, then chordwise to the TE.

CRITICAL: Do NOT stretch the membrane spanwise. Allow it to lie naturally — it should be just wrinkle-free, not taut.

Do NOT bond the inter-rib membrane spans to anything. Only the rib faces, LE strip, and TE strip receive adhesive. The membrane must be free to billow between ribs under aerodynamic pressure.

STEP 8 — Bond Trailing Edge Strip and Trim Membrane

Before bonding the TE strip, fold the membrane over the trailing edge of the last rib. Apply contact adhesive to the TE strip faces and the membrane in that zone.

Press the 0.5mm × 4mm CFRP TE strip onto the trailing edges of all ribs, sandwiching the membrane against them. Press firmly along the full span.

Using the planform template as a guide, trim the membrane to the wing planform shape with sharp scissors. Cut cleanly along the template edge.

Trim the leading edge membrane flush with the front of the LE fairing strip.

Trim the tip edge perpendicular to the span axis.

The wing planform is now defined by the trimmed membrane edge, not by the rib chord length. Ribs that extend past the membrane edge are fine — they are simply internal formers.

STEP 9 — Apply Lower Membrane

Repeat Steps 7 and 8 for the lower surface.

Bond the lower membrane to the lower faces of all ribs, the lower face of the TE strip, and the underside of the LE fairing strip at the leading edge.

The lower membrane should be bonded very slightly tighter than the upper — it should lie nearly flat when the wing is held horizontal.

Trim to the same planform template as the upper membrane.

CHECK: Hold the wing at the root and blow gently from the LE at approximately 1 m/s. The upper membrane should visibly billow 3-5mm above the rib profile. The lower membrane should remain nearly flat. If the upper does not billow: the tension is too high — carefully slit the bond at one or two inter-rib stations and re-bond with less tension.

6.7 Film Covert Flaps

STEP 10 — Cut and Bond Film Flap Strips

Cut three strips of 20-micron ripstop nylon (or Mylar) for each wing.

Strip dimensions: width = 9.60mm (flap chord), length = span coverage from 15% to 85% semi-span measured along the actual wing.

Apply contact adhesive to the FORWARD EDGE ONLY of each strip — the first 2mm of the strip width. Also apply to the flap ledge recesses on the rib upper surfaces.

Allow 60 seconds to become tacky.

Press the forward edge of each film strip into its corresponding ledge recess. Row 1 at 45mm, Row 2 at 63mm, Row 3 at 81mm from the LE.

The rear portion of each strip (7-8mm) must be COMPLETELY FREE. Do not apply any adhesive rearward of the ledge. Do not press the rear edge down.

CHECK: Lift each strip from its rear edge with a fingernail. It should hinge freely from the bond line. If it does not lift freely the rear is inadvertently bonded — carefully peel back and re-do.

The bond line IS the hinge. No piano wire, no spring, no magnet.

6.8 Wing Fences and End Plate

STEP 11 — Bond Wing Fences

Cut fence strips from 0.2mm CFRP sheet: height = 3% of local chord, chord extent = 10% to 72% of local chord.

Fence positions: W1 at 45% semi-span; W2 at 40%; W3 at 30% AND 60%; W4 at 50%.

Apply thin CA to the fence base and press onto the upper membrane surface perpendicular to the chord plane.

Verify perpendicularity within 2° using a small set square.

Hold for 30 seconds. Do not disturb for 5 minutes.

STEP 12 — Bond End Plate

The end plate is a NACA 0008 symmetric section, cut from 0.5mm CFRP.

Dimensions: approximately 15% of wing plan area. Height above chord plane: 8% semi-span. Height below: 8% semi-span.

The upper plate face is CANTED 15° outboard (not vertical). The lower plate is vertical.

Bond to the tip rib with CA and one M2 bolt through a pre-drilled hole.

Verify the 15° outboard cant on the upper plate with a protractor.

6.9 Root Attachment Plate

STEP 13 — Fit Root Attachment Plate

Bond the 2mm CFRP root attachment plate to the root rib and LE spar using Araldite 2011 structural epoxy.

The plate has a 2.10mm clearance hole for the LE spar and two M3 bolt holes for attachment to the pitch axis of the wing root mechanism.

Clamp for 60 minutes minimum cure. Do not load for 24 hours.

Bolt the root rib to the spar through the plate using M3 screws with thread-locker (Loctite 243).

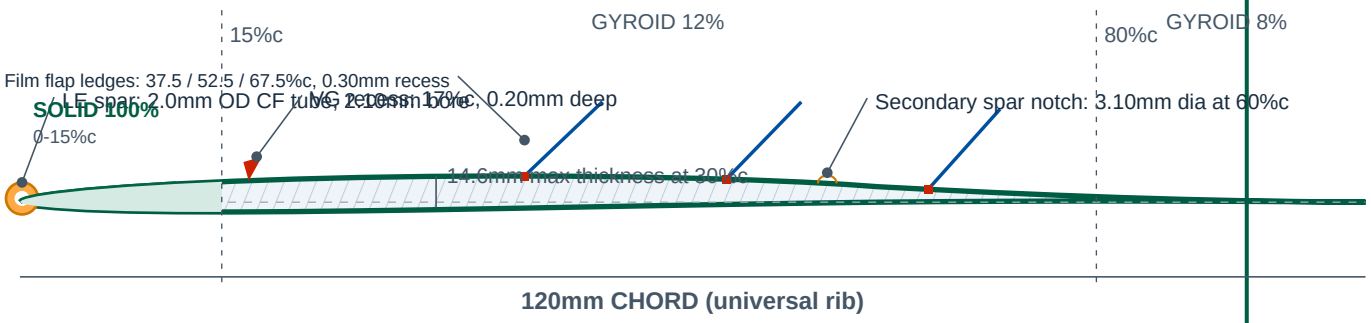
6.10 Final Quality Inspection

Parameter	Acceptance Criterion
Wing planform	Matches template within 2mm on all edges
Trailing edge	Sharp, less than 0.5mm radius, no step at membrane joint
Natural frequency	1.8-2.2 Hz measured and recorded

Parameter	Acceptance Criterion
LE bore	2.0mm spar slides freely; 2.2mm does not pass
Membrane billow	Upper surface billows 3-5mm at 1 m/s gentle airflow from LE
Film flap Row 1	Lifts freely when blown at approximately 15° AoA, 1 m/s
Film flap Row 2	Lifts freely at approximately 20° AoA
Film flap Row 3	Lifts freely at approximately 28° AoA
Film flap rear edges	Completely free throughout their length. No accidental bonding.
VG bonding	All VGs firmly bonded. No peeling at leading edge of any VG.
Wing fence	Perpendicular to chord plane within 2°
End plate cant	Upper plate 15° outboard within 2°
Root plate	Flat within 0.3mm. Bolts secure with thread-locker applied.
Wing mass	W1: 22-28g. W2: 20-26g. W3: 21-27g. W4: 18-24g.
Mirror symmetry	Left and right wings at same position within 2g mass

A wing that passes all quality checks is ready for installation into the wing root mechanism. A wing that fails any check should be repaired or remade before installation. The natural frequency check is the most important single criterion — it determines aerodynamic efficiency more than any other measurable parameter.

End of Wing Manufacturing Document. Blueprint schematic of S1223 rib cross-section follows.



KEY:

Solid LW-PLA nose zone (0-15%c) · Gyroid infill zones · Carbon fibre spar · LE fairing strip (0.30mm) · Film CF flap ledges (3 positions) · VG recess position · Adhesive bond line

MCO-1D — S1223 RIB CROSS-SECTION — UNIVERSAL 120mm DESIGN

LW-PLA gyroid 12% · Solid nose zone · Three film flap ledges · VG recess · LE and secondary spar bores

Schematic — not to scale

ADDENDUM R20 — AS-BUILT SPECIFICATION UPDATE

This addendum supersedes specific numerical values and adds chapters missing from the original Wing Manufacturing Document (May 2026, 24 pages). Where this addendum contradicts the original document, this addendum governs. All other original content remains valid.

Section	Topic	Change from original
R20-1	Wing Spans — Final As-Built Specification	All four spans revised
R20-2	Rib Count and Spacing — Updated	Spans changed rib counts
R20-3	Servo Specification — AGFRC IB53BHU	Original had placeholder spec
R20-4	Root Attachment Interface	New chapter — not in original
R20-5	Rope Drive and Sector Wheel Interface	New chapter — not in original
R20-6	ODrive S1 Motor Mount and Integration	New chapter — not in original
R20-7	Per-Wing Build Sequence (W1–W4)	New chapter — not in original
R20-8	Updated Mass Summary	Reflects final spec
R20-9	Updated BOM — Wing Hardware Only	Corrected to final spec
R20-10	Cross-References to Other Documents	Links to Comm Manual v20

■ **SERVO PROGRAMMING MUST BE DONE BEFORE INSTALLATION. Programme all 16 flight servos to 170° maximum travel using the AGF-SPV3 programmer before installing in the aircraft. The ADC measuring servo position saturates at 180° — 170° provides the 10° safety margin. See R20-3 and Commissioning Manual v20 Assembly Card Phase A.**

R20-1 WING SPANS — FINAL SPECIFICATION

Supersedes STEP PREP-2 in original document

The wing spans listed in STEP PREP-2 of the original document (W1=550mm, W2=495mm, W3=578mm, W4=523mm) are the early design values. The final MCO-1D specification uses the following semi-spans. These are the values to use for spar cutting.

Wing pair	Position	Original span (May 2026)	FINAL span (Rev 20)	Change
W1 (pair 1)	Forward / largest	550 mm	772 mm	+222 mm
W2 (pair 2)	Second from front	495 mm	749 mm	+254 mm
W3 (pair 3)	Third from front	578 mm	735 mm	+157 mm
W4 (pair 4)	Rear / smallest	523 mm	712 mm	+189 mm

All spans are **semi-spans** (root to tip). The spar cut length equals the semi-span. The secondary spar rod is cut to semi-span minus 20mm (10mm clear each end) as specified in the original document.

ROOT CHORD — UNCHANGED

The 120mm universal chord rib remains correct. The root chord as seen at the bolt circle attachment is 230–250mm (measured along the swept leading edge) tapering to 165mm at the innermost rib over the first 180mm of span. This is unchanged from the original design.

■ The larger spans significantly increase the moment arm at the root attachment. Ensure all root rib bonds are given full cure time (24h for CA) before any load testing. The rope drive applies the primary load at the root.

R20-2 RIB COUNT AND SPACING — UPDATED

Revised from original 110 ribs total

With the revised semi-spans, the original 110-rib count and print run are no longer correct. The following table gives the correct rib counts for the final specification. Rib spacing remains 25–30mm throughout.

Wing pair	Semi-span	Rib spacing	Ribs per wing	Ribs per pair (×2)
W1	772 mm	28 mm	28	56
W2	749 mm	28 mm	27	54
W3	735 mm	27 mm	27	54
W4	712 mm	27 mm	26	52
Total (8 wings)	—	—	108	216

Print run: 216 ribs at 0.9g average = approximately 195g total LW-PLA. At 15–22 hours for 110 ribs, the revised print run is approximately 30–44 hours total. Plan for two print sessions on a typical 220mm × 220mm bed. All 216 ribs are identical — the universal 120mm rib SolidWorks model is unchanged from the original document.

Note: Add 10–15% overage to the print run for first-time builds. Ribs occasionally delaminate during membrane bonding if the CA cure was rushed. Having spares is faster than reprinting mid-assembly.

R20-3 SERVO SPECIFICATION — AGFRC IB53BHU

Final servo specification for all 16 wing positions

The original wing manufacturing document did not specify a servo part number. The final MCO-1D specification uses the AGFRC IB53BHU throughout. One servo is installed per wing at the root, driving the sector wheel for sweep/AoA control. Two servos per pair (port + starboard) = 16 total.

Parameter	Specification
Part number	AGFRC IB53BHU
Programmer required	AGF-SPV3 (one per build — do not skip)
Travel before programming	Default ±180° — DO NOT INSTALL AT DEFAULT
Travel after programming	170° maximum (MUST be set before installation)
Why 170°?	ADC measuring servo position saturates at 180° — 10° margin
Torque	≥ 8 kg·cm at 7.4V operating voltage
Speed	≥ 0.11s / 60° at 7.4V
Connector	Standard JR / Futaba compatible
Quantity	18 total: 16 flight + 2 spares (order and programme all 18)

■ **PROGRAMME ALL 18 SERVOS TO 170° BEFORE INSTALLING ANY. Using the AGF-SPV3 programmer: connect each servo individually, set endpoint to 170°, confirm, disconnect. Label each servo after programming. A servo installed at factory default will drive the sector wheel past its mechanical stop on the first powered test.**

LABELLING CONVENTION

Label each servo with its wing assignment after programming and before installation:

Label	Wing	ODrive node_id	Position
W1p	Wing 1 port (left)	0	Forward pair, port side
W1s	Wing 1 starboard (right)	1	Forward pair, starboard side
W2p	Wing 2 port	2	Second pair, port side
W2s	Wing 2 starboard	3	Second pair, starboard side
W3p	Wing 3 port	4	Third pair, port side
W3s	Wing 3 starboard	5	Third pair, starboard side
W4p	Wing 4 port	6	Rear pair, port side
W4s	Wing 4 starboard	7	Rear pair, starboard side

R20-4 ROOT ATTACHMENT INTERFACE

New chapter — not in original May 2026 document

The original document ended at the completed wing. This chapter covers the root attachment — how the completed wing connects to the fuselage frame and sector wheel drive mechanism.

ROOT RIB PREPARATION

The root rib is the innermost rib — nearest to the fuselage centreline. Unlike all other ribs (which are simply CA-bonded to the spar), the root rib requires additional preparation before bonding:

- Sand the root face (the face that will contact the attachment plate) with 120-grit sandpaper to create a mechanical key for epoxy bonding.
- Drill or pre-form two 2.5mm clearance holes through the root rib body, perpendicular to the chord plane, for M2.5 attachment bolts. Positions: 20mm and 80mm from the leading edge.
- Bond the root rib to the main spar with 5-minute epoxy (not CA). CA is adequate for all other ribs but the root rib carries the full drive load — epoxy provides the shear strength needed.
- Allow 24h full cure before applying any drive load.

ATTACHMENT PLATE

The attachment plate is a 3mm CFRP plate, CNC-cut or hand-cut, with the following features:

- Two M2.5 through-holes matching the root rib bolt pattern
- One M3 clearance hole for the sector wheel pivot bolt
- Positive stop for the sector wheel arc limit (mechanical end-stop)
- Bond and bolt to root rib: thin layer of structural epoxy on mating face, M2.5 bolts with threadlocker

SWEEP SERVO MOUNT

The AGFRC IB53BHU servo mounts at the root of each wing on the attachment plate. The servo output horn connects to the sector wheel via the rope drive. Servo axis must be perpendicular to the wing chord plane.

- Servo mount: aluminium bracket or 3D-printed LW-PLA bracket
- Fastening: M2 screws with washer and threadlocker
- Clearance: ensure the servo horn clears the sector wheel arc through the full 170° programmed travel

Note: The servo does NOT directly drive the wing. It drives the sector wheel via the rope. The rope tension must be confirmed before any powered test — see mco_odrive_setup.py soft-stop configuration.

R20-5 ROPE DRIVE AND SECTOR WHEEL INTERFACE

New chapter — not in original May 2026 document

The rope drive is the primary structural and control interface between the ODrive S1 motor and the wing. It is also the primary fatigue failure mode. This chapter covers rope selection, installation, tensioning, and the mandatory fatigue test.

SECTOR WHEEL

Each wing is driven through a sector wheel — a partial circular gear segment with a groove for the rope. The sector wheel converts the motor's rotational motion into wing flapping and sweep motion.

Parameter	Value	Notes
Sector arc	C2 to C3 degrees (measure actual)	Enter in mco_build_measurements.py as C2, C3
Groove width	1.2mm	Matches rope diameter
Groove depth	0.8mm	Rope must seat fully — verify before each flight
Gear ratio	C1 (measure actual)	Count teeth or measure arc vs motor turns
Material	6061-T6 aluminium or PETG-CF print	Aluminium preferred for fatigue

ROPE SPECIFICATION

Parameter	Specification
Material	Spectra / Dyneema braided, 0.8–1.0mm diameter
Breaking strength	≥10× peak drive load (verify with load cell)
Fatigue test	MANDATORY before first powered test — see below
ROPE_LIFE_I2_CYCLES	Must be measured — no default accepted
Inspection interval	Check for fraying at reversal points every flight

■ **ROPE FATIGUE TEST IS MANDATORY BEFORE ANY POWERED TEST. Drive the rope at rated motor current until failure. Record I² × cycles at failure. Enter this value as ROPE_LIFE_I2_CYCLES in mco_certification_tools.py. The rope monitor uses this value to predict remaining life. There is NO safe default — every rope batch is different.**

ROPE INSTALLATION

- Thread rope through the sector wheel groove, starting from the neutral position.
- Route to the wing root attachment point. The rope must not contact any edge except the groove itself.
- Tension: the rope should have zero slack at the neutral position but must not pre-load the servo horn. Check that the servo horn moves freely with rope attached.
- Knot or clamp the rope at both ends. A double-wrapped hitching knot with CA reinforcement is acceptable. No slipping permitted under rated load.
- Run the motor through the full sector arc under software control before any flight load. Confirm rope stays in groove throughout.
- BEFORE EACH FLIGHT: visually inspect rope at both reversal points for fraying, kinking, or groove mis-seating.

R20-6 ODRIVE S1 MOTOR MOUNT AND INTEGRATION

New chapter — not in original May 2026 document

Each wing is driven by one GoolRC 5010-14 360KV brushless motor controlled by a dedicated ODrive S1 controller. The eight controllers form a CAN-FD bus daisy-chain.

MOTOR MOUNT

Parameter	Value
Motor	GoolRC 5010-14 360KV — one per wing
Controller	ODrive S1 — ONE PER MOTOR (8 total)
Node assignment	node_id 0 = W1p, 1 = W1s, 2 = W2p ... 7 = W4s
CAN-FD bus	Daisy-chain all 8 boards. 120Ω termination resistors at both ends.
Encoder	On-board absolute magnetic encoder — no additional hardware
Encoder magnet	8mm × 2mm N52 neodymium disc, press-fit to motor shaft

ENCODER MAGNET INSTALLATION

- Press-fit the encoder magnet to the motor shaft end. Magnet face must be within 0.5–1.5mm of the PCB encoder surface. Use the ODrive S1 LED feedback to confirm the magnet is reading correctly.
- After installation: connect ODrive, power on, run `odrv0.axis0.encoder.pos_estimate` and confirm it reads a value and changes smoothly as you turn the shaft by hand.
- Run mco_odrive_setup.py for each motor. This sets: current limits, velocity limits, soft-stop sector arc limits, and encoder offset. Run it in order from node_id 0 to 7.

SOFT-STOP CONFIGURATION (CRITICAL)

The soft-stop zones prevent the sector wheel from overrunning its mechanical limits. mco_odrive_setup.py sets these automatically once C2 and C3 are measured and entered. After setup, verify with:

```
GS: soft_stop_status # must show PASS for all 8 wings before any flight
```

CAN-FD BUS WIRING

- Use twisted pair for the CAN_H and CAN_L lines throughout.
- Maximum segment length between boards: 300mm recommended.
- Both physical ends of the bus chain (controller 0 and controller 7) must have 120Ω termination resistors installed on the CAN connector.
- Each ODrive S1 has a solder pad for the termination resistor — install on node_id 0 and node_id 7 only.
- Test the bus: boot the Pi, run mco_can_setup.sh, confirm all 8 nodes respond to a broadcast message.

Note: The ODrive S1 uses CAN-FD (not classic CAN). The Raspberry Pi CAN hat must be configured for CAN-FD bitrate matching. See mco_can_setup.sh for the correct configuration.

R20-7 PER-WING BUILD SEQUENCE (W1–W4)

Build order and pair-specific notes

Build all four pairs in the sequence: W1 first, W4 last. W1 carries the highest structural load (largest span, most forward, highest root moment arm) and is most critical to get right. Validate W1 fully before starting W2.

Pair	Semi-span	Dihedral	Rib count	Specific notes
W1	772 mm	3°	28 per wing	Largest pair. Highest root load. Build first. Validate rope fatigue on W1 before oth
W2	749 mm	5°	27 per wing	Second largest. Mirror W1 jig settings for dihedral after measuring actual.
W3	735 mm	7°	27 per wing	Highest dihedral of the four pairs. Verify dihedral jig angle carefully.
W4	712 mm	4°	26 per wing	Smallest pair. Lowest mass. Build last. Pairs must balance port vs starboard to w

PORT/STARBOARD MIRRORING

Each pair consists of one port wing and one starboard wing. They are mirror images. Build all port wings first with the jig in the port orientation, then flip the jig for all starboard wings. Do not mix port and starboard ribs — the VG alignment dimples are on the correct faces only when the rib is oriented correctly.

BALANCE CHECK

- Weigh each completed wing (including servo, attachment plate, but excluding rope) on a jeweller's scale (0.01g resolution).
- Port and starboard of the same pair must be within 2g of each other. If more than 2g difference, add balance mass (small lead shot or epoxy dab) to the lighter wing at the tip, tracking the mass added.
- Record actual mass of each wing in mco_build_measurements.py. These values are used by cg_management_tick.

R20-8 UPDATED MASS SUMMARY AND R20-9 BOM UPDATE

Corrected to final Rev 20 specification

R20-8 — WING MASS SUMMARY (ALL 8 WINGS)

Item	Spec	Mass (estimate)	Notes
LW-PLA ribs (216 total)	120mm chord, 2mm thick	~195g	At 0.9g per rib average
Carbon fibre spars	2.0mm OD / 1.0mm ID CF tube	~35g	LE spar all 8 wings
Secondary spar rods	1.5mm CFRP rod	~12g	One per wing
Ripstop nylon membrane	30D, 4g/m²	~35g	Upper + lower all 8 wings
Film covert flaps	20-micron PET film	~5g	3 rows per wing
LE fairing strips	0.3mm CFRP strip, 6mm wide	~8g	All 8 wings
TE strips	0.5mm CFRP strip	~10g	All 8 wings
VG vane arrays	0.3mm CFRP sheet, laser cut	~6g	~24 pairs per wing
Adhesives and sealant	Contact cement, CA, epoxy	~15g	Estimated
Wing hardware subtotal	—	~321g	Structural wing only
Servos (16× AGFRC IB53BHU)	~33g each	~528g	16 flight servos
Attachment plates + brackets	3mm CFRP, M2.5 bolts	~48g	16 per aircraft
Encoder magnets (8×)	8mm×2mm N52	~8g	
TOTAL WING GROUP	—	~905g	Approximately 26% of MTOW

R20-9 — BILL OF MATERIALS UPDATE

Items changed from the original BOM (May 2026) are marked ▲. All prices USD as of August 2026. Verify before ordering.

Item ▲=changed	Spec	Qty	Est. unit USD
▲ LW-PLA filament	Colorfabb LW-PLA or Spectrum Ultra Foam, 750g	2 × 750g	~$35
S1223 coordinate file	UIUC Airfoil Database s1223.dat (free)	1	—
▲ LE spar CF tube	2.0mm OD / 1.0mm ID pultruded, 1m lengths	12 lengths	~$6
Secondary spar CFRP rod	1.5mm CFRP rod, 1m lengths	10 lengths	~$4
Ripstop nylon (upper)	30 denier, 4g/m², 1m × 1m	2 pieces	~$12
Ripstop nylon (lower)	30 denier, 4g/m², 1m × 1m	2 pieces	~$12
Film covert flaps	20-micron PET film, 300mm × 500mm sheet	1	~$8
LE fairing CFRP strip	0.3mm × 6mm strip, 1m	4 lengths	~$5
TE CFRP strip	0.5mm × 8mm strip, 1m	4 lengths	~$6
VG CFRP sheet	0.3mm, 150mm × 150mm	2 sheets	~$8
▲ Flight servos	AGFRC IB53BHU — purchase and programme ALL 18	18	~$28
▲ Servo programmer	AGF-SPV3 — required to set 170° limit	1	~$22
3M 9460PC film adhesive	VG bonding	1 small roll	~$15
Contact cement	UHU Por or Bostik Stick-2	2 tubes	~$10
CA thin adhesive	Spar-to-rib bonds	3 × 20g bottles	~$8
5-minute epoxy	Root rib bonds only	1 syringe pair	~$6
Root attach plates	3mm CFRP sheet 100mm×100mm, CNC-cut	16 pieces	~$5

M2.5 bolts and nuts	SS, 8mm length	1 pack (50)	~$6
Encoder magnets	8mm×2mm N52 neodymium disc	1 pack (10)	~$8

R20-10 CROSS-REFERENCES TO OTHER DOCUMENTS

Navigation guide for the full MCO-1D document suite

The MCO-1D documentation suite consists of the following documents. This addendum covers wing build only. Other build aspects are covered in the documents listed below.

Document	What it covers	Relevant for wing build
Commissioning Manual v20 (51pp)	Complete assembly sequence Phases A–F, software setup, flight testing	Phase A servo programming (TC-01 through TC-04) during wing install; Phase B
Assembly Card (Field Cards v20, 5pp)	One-page laminated quick reference for assembly sequence	Carry during build. Phases A–C must complete before any po
Bill of Materials v18 (8pp)	Full aircraft BOM including electronics, fuselage and tail	Wing hardware BOM in this addendum (R20-9) supersedes th
Software Reference v20 (8pp)	GS commands, tick order, knowledge files	After first flight: mco_build_measurements.py entries feed dire
Build Measurement Sheet (2pp)	Blanks for all calibration constants	Fill in during build: A1 (MTOW), C1/C2/C3 (sector wheel), K_F

DOCUMENT VERSION SUMMARY

Document	Current version	Date
Wing Manufacturing Document + Addendum R20	Original + R20	Aug 2026
Commissioning Manual	v20 (51 pages)	Aug 2026
Field Cards	v20 (5 cards, laminated A5)	Aug 2026
Software Reference	v20 (8 pages)	Aug 2026
Bill of Materials	v18 (8 pages; wing section: see R20-9)	Aug 2026
Build Measurement Sheet	Rev A (2 pages)	Jun 2026